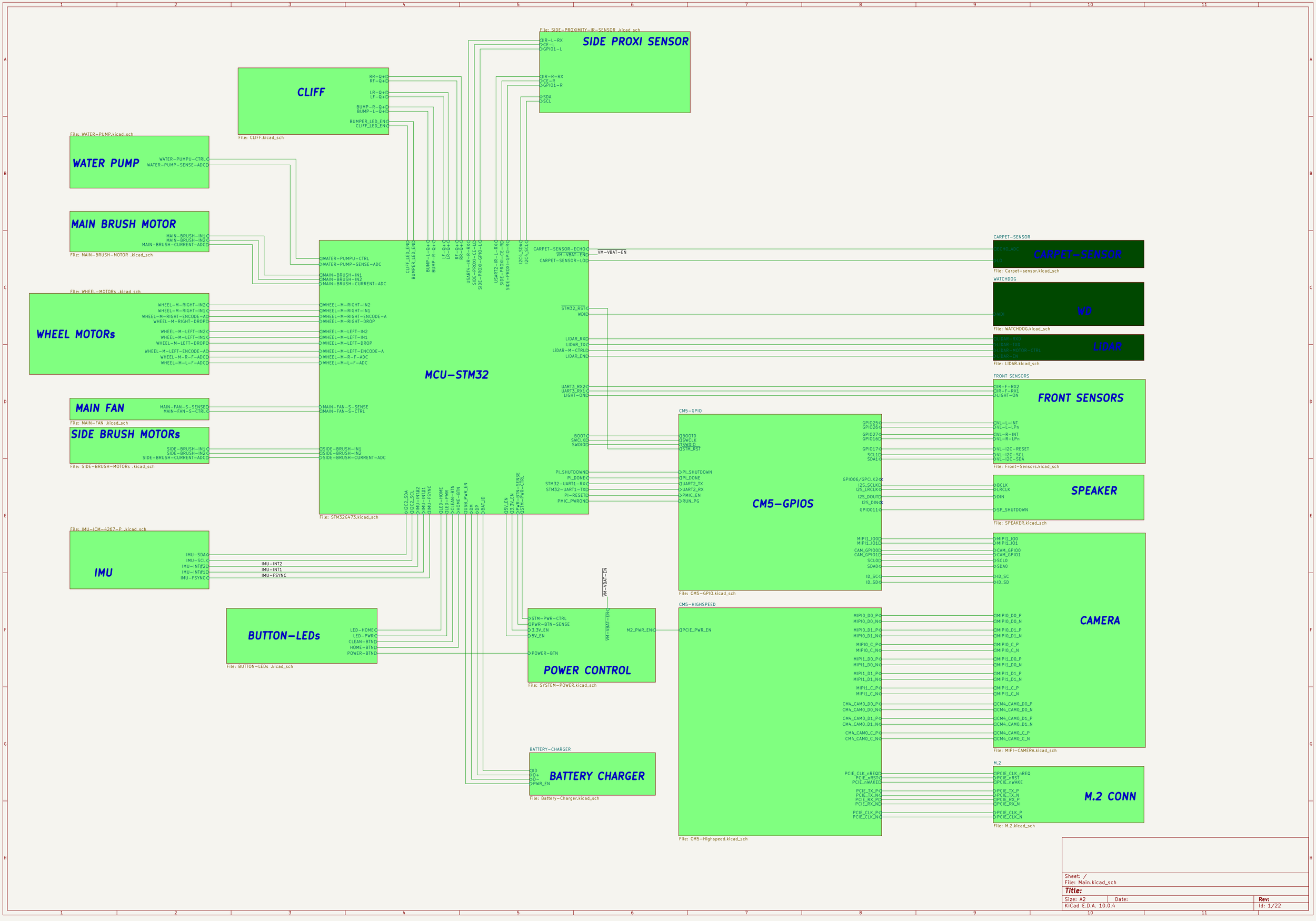
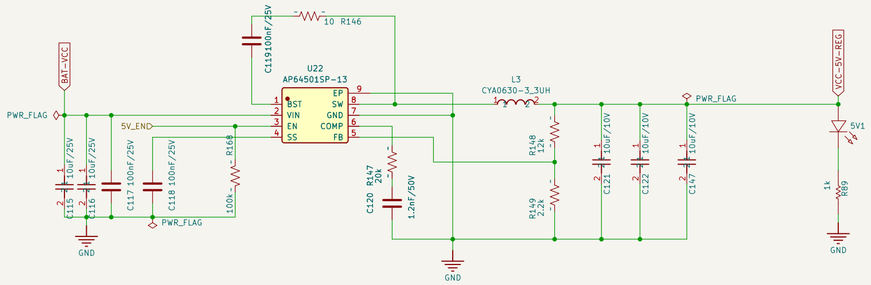
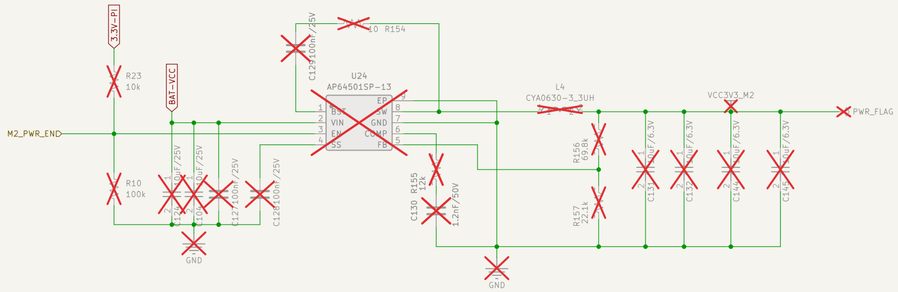
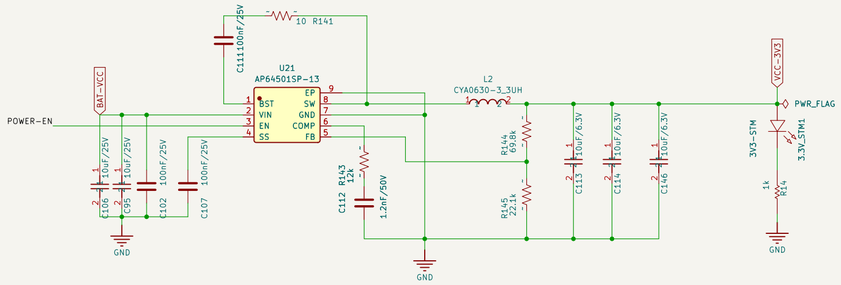
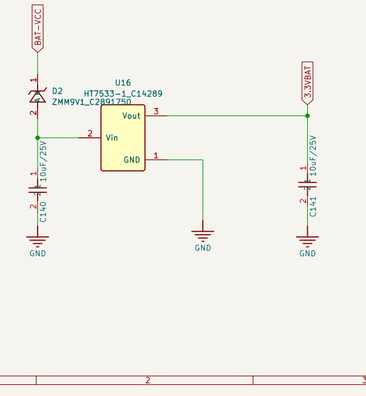

File: SIDE-PROXIMITY-IR-SENSOR .kicad_sch
SIDE PROXI SENSOR
IR-L-RX
CE-L
GPIO1-L
IR-R-RX
CE-R
GPIO1-R
SDA
SCL
CLIFF
RR-Q+D
RF-Q+D
LR-Q+D
LF-Q+D
BUMP-R-Q+D
BUMP-L-Q+D
BUMPER_LED_EN
CLIFF_LED_EN
File: CLIFF.kicad_sch
File: WATER-PUMP.kicad_sch
WATER PUMP
WATER-PUMPU-CTRL
WATER-PUMP-SENSE-ADCD
MAIN BRUSH MOTOR
MAIN-BRUSH-IN1
MAIN-BRUSH-IN2
MAIN-BRUSH-CURRENT-ADCD
File: MAIN-BRUSH-MOTOR .kicad_sch
File: WHEEL-MOTORs .kicad_sch
WHEEL MOTORs
WHEEL-M-RIGHT-IN2
WHEEL-M-RIGHT-IN1
WHEEL-M-RIGHT-ENCODE-AD
WHEEL-M-RIGHT-DROPD
WHEEL-M-LEFT-IN2
WHEEL-M-LEFT-IN1
WHEEL-M-LEFT-DROPD
WHEEL-M-LEFT-ENCODE-AD
WHEEL-M-R-F-ADCD
WHEEL-M-L-F-ADCD
MCU-STM32
WATER-PUMPU-CTRL
WATER-PUMP-SENSE-ADC
MAIN-BRUSH-IN1
MAIN-BRUSH-IN2
MAIN-BRUSH-CURRENT-ADC
WHEEL-M-RIGHT-IN2
WHEEL-M-RIGHT-IN1
WHEEL-M-RIGHT-ENCODE-A
WHEEL-M-RIGHT-DROP
WHEEL-M-LEFT-IN2
WHEEL-M-LEFT-IN1
WHEEL-M-LEFT-DROP
WHEEL-M-LEFT-ENCODE-A
WHEEL-M-R-F-ADC
WHEEL-M-L-F-ADC
MAIN FAN
MAIN-FAN-S-SENSED
MAIN-FAN-S-CTRLG
File: MAIN-FAN .kicad_sch
MAIN-FAN-S-SENSE
MAIN-FAN-S-CTRL
SIDE BRUSH MOTORs
SIDE-BRUSH-IN1
SIDE-BRUSH-IN2
SIDE-BRUSH-CURRENT-ADCD
File: SIDE-BRUSH-MOTORs .kicad_sch
SIDE-BRUSH-IN1
SIDE-BRUSH-IN2
SIDE-BRUSH-CURRENT-ADC
File: STM32G473.kicad_sch
CLIFF_LED_EN
BUMPER_LED_EN
BUMP-L-Q+C
BUMP-R-Q+C
LF-Q+C
LR-Q+C
RF-Q+C
RR-Q+C
USART4-IR-R-RX
SIDE-PROXI-CE-L
SIDE-PROXI-GPIO-L
USART2-IR-L-RX
SIDE-PROXI-CE-R
SIDE-PROXI-GPIO-R
I2C4_SDA
I2C4_SCL
I2C2_SDA
I2C2_SCL
IMU-INT#2
IMU-INT#1
IMU-FSYNC
LED-HOME
LED-PWR
CLEAN-BTN
HOME-BTN
POWER-BTN
USB_PWR_EN
DM
DP
BAT_ID
5V_EN
3.3V_EN
PWR-BTN-SENSE
STM-PWR-CTRL
CARPET-SENSOR-ECHO
VM-VBAT-END
CARPET-SENSOR-LOD
VM-VBAT-EN
STM32_RST
WDI
LIDAR_RX
LIDAR_TX
LIDAR-M-CTRL
LIDAR_END
UART3_RX2
UART3_RX1
LIGHT-ON
BOOT
SWCLK
SWDIO
STM_RST
PI_SHUTDOWN
PI_DONE
STM32-UART1-RX
STM32-UART1-TX
PI-RESET
PMIC_PWROND
File: IMU-ICM-4267-P .kicad_sch
IMU
IMU-SDA
IMU-SCL
IMU-INT#2D
IMU-INT#1D
IMU-FSYNC
IMU-INT2
IMU-INT1
IMU-FSYNC
BUTTON-LEDs
LED-HOME
LED-PWR
CLEAN-BTND
HOME-BTND
POWER-BTND
File: BUTTON-LEDs .kicad_sch
STM-PWR-CTRL
PWR-BTN-SENSE
3.3V_EN
5V_EN
POWER-BTN
VM-VBAT-EN
M2_PWR_EN
POWER CONTROL
File: SYSTEM-POWER.kicad_sch
BATTERY-CHARGER
ID
D+
D-
PWR_EN
BATTERY CHARGER
File: Battery-Charger.kicad_sch
CM5-GPIO
BOOT0
SWCLK
SWDIO
STM_RST
PI_SHUTDOWN
PI_DONE
UART2_TX
UART2_RX
PMIC_EN
RUN_PG
GPIO25
GPIO26
GPIO27
GPIO16
GPIO17
SCL1
SDA1
GPIO06/GPCLK2
I2S_SCLK
I2S_LRCLK
I2S_DOUT
I2S_DIN
GPIO011
MIPI1_IO0
MIPI1_IO1
CAM_GPIO0
CAM_GPIO1
SCL0
SDA0
ID_SC
ID_SD
CM5-GPIOS
File: CM5-GPIO.kicad_sch
CM5-HIGHSPEED
PCIE_PWR_EN
MIPI0_D0_P
MIPI0_D0_N
MIPI0_D1_P
MIPI0_D1_N
MIPI0_C_P
MIPI0_C_N
MIPI1_D0_P
MIPI1_D0_N
MIPI1_D1_P
MIPI1_D1_N
MIPI1_C_P
MIPI1_C_N
CM4_CAM0_D0_P
CM4_CAM0_D0_N
CM4_CAM0_D1_P
CM4_CAM0_D1_N
CM4_CAM0_C_P
CM4_CAM0_C_N
PCIE_CLK_nREQ
PCIE_nRST
PCIE_nWAKE
PCIE_TX_P
PCIE_TX_N
PCIE_RX_P
PCIE_RX_N
PCIE_CLK_P
PCIE_CLK_N
File: CM5-Highspeed.kicad_sch
CARPET-SENSOR
ECHO_ADC
CLO
CARPET-SENSOR
File: Carpet-sensor.kicad_sch
WATCHDOG
WDI
WD
File: WATCHDOG.kicad_sch
LIDAR-RXD
LIDAR-TXD
LIDAR-MOTOR-CTRL
LIDAR-EN
LIDAR
File: LIDAR.kicad_sch
FRONT SENSORS
IR-F-RX2
IR-F-RX1
LIGHT-ON
VL-L-INT
VL-L-LPn
VL-R-INT
VL-R-LPn
VL-I2C-RESET
VL-I2C-SCL
VL-I2C-SDA
File: Front-Sensors.kicad_sch
BCLK
LRCLK
DIN
SP_SHUTDOWN
SPEAKER
File: SPEAKER.kicad_sch
MIPI0_D0_P
MIPI0_D0_N
MIPI0_D1_P
MIPI0_D1_N
MIPI0_C_P
MIPI0_C_N
MIPI1_D0_P
MIPI1_D0_N
MIPI1_D1_P
MIPI1_D1_N
MIPI1_C_P
MIPI1_C_N
CM4_CAM0_D0_P
CM4_CAM0_D0_N
CM4_CAM0_D1_P
CM4_CAM0_D1_N
CM4_CAM0_C_P
CM4_CAM0_C_N
CAMERA
File: MIPI-CAMERA.kicad_sch
M.2
PCIE_CLK_nREQ
PCIE_nRST
PCIE_nWAKE
PCIE_TX_P
PCIE_TX_N
PCIE_RX_P
PCIE_RX_N
PCIE_CLK_P
PCIE_CLK_N
M.2 CONN
File: M.2.kicad_sch
Sheet: /
File: Main.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/22

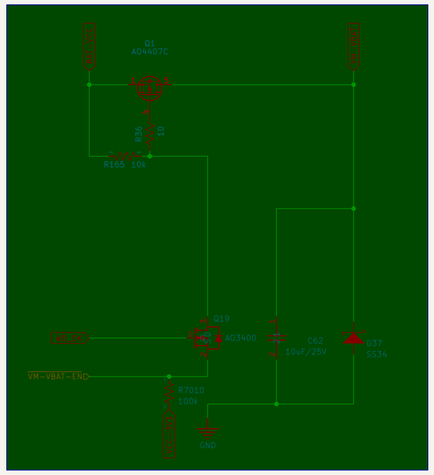
# POWER MANAGMENT



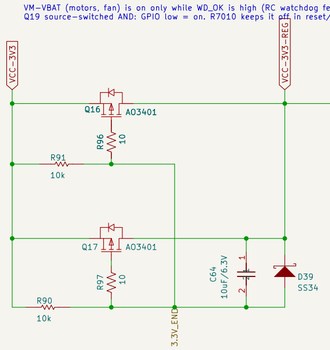



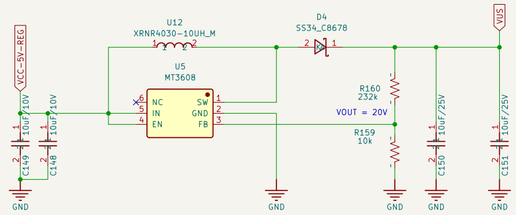



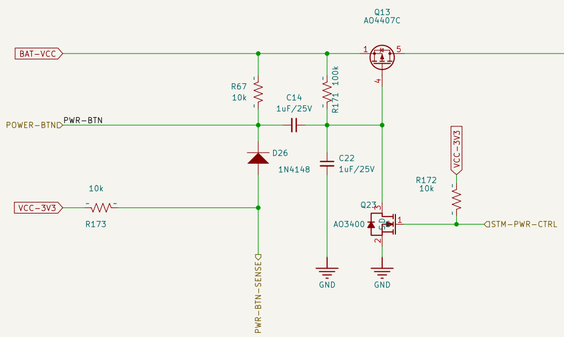

TODO Check VM-VBAT max current vs Q1 rating

Old VM-5V switch removed: the LiDAR (VM-5V-LIDAR) and the water pump (VM-5V-WATER-PUMP) now have their own WD_OK-gated switches on their sheets.
No soft ramp.

VM-VBAT (motors, fan) is on only while WD_OK is high (RC watchdog fed) AND VM-VBAT-EN is low.
Q19 source-switched AND: GPIO low = on. R7010 keeps it off in reset/flash. AO4407C: Vgs +/-25 V, no gate divider needed.

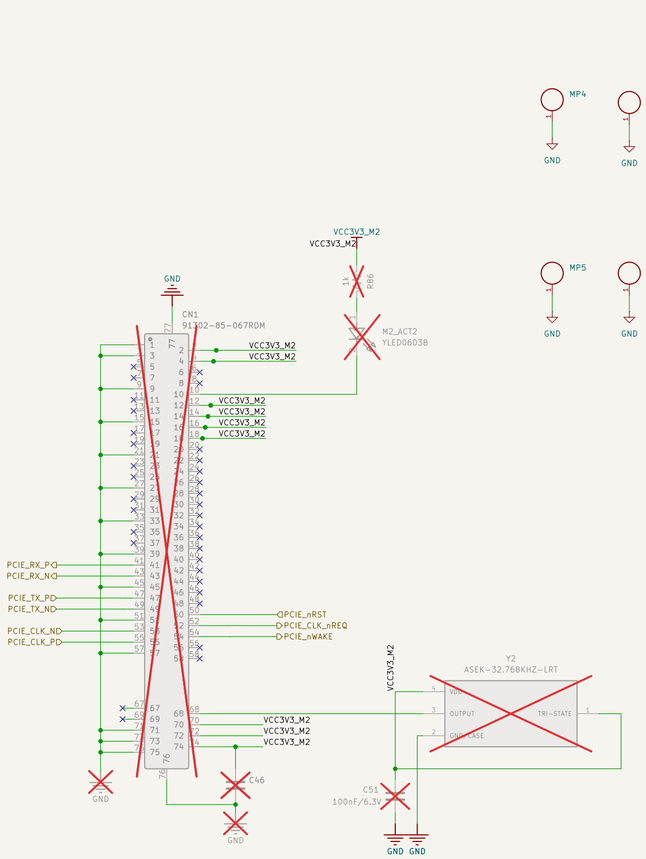


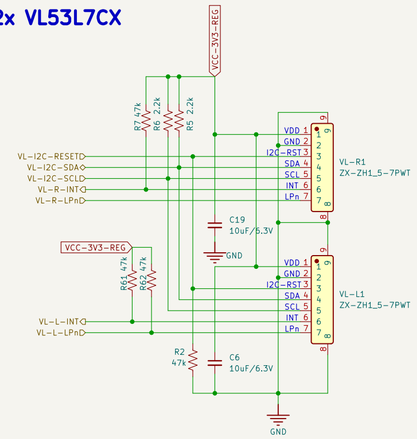

STM-PWR-CTRL need to use for both EN and OFF from STM32 GPIO pin.
1- Hold EN to HIGH after buttone pressed
2- OFF board

| | | |
|---|---|---|
| Sheet: /POWER/ | | |
| File: SYSTEM-POWER.kicad_sch | | |
| **Title:** | | |
| Size: A2 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | | Id: 2/22 |

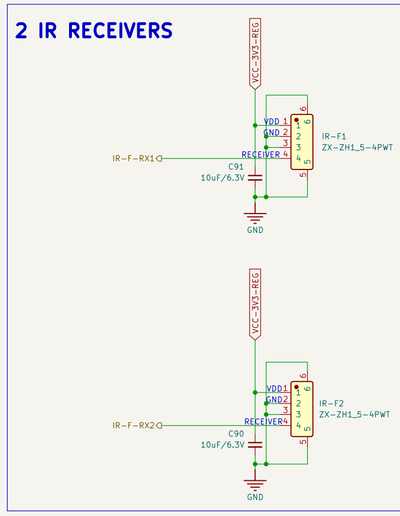



| | |
|---|---|
| Sheet: /M.2/ | |
| File: M.2.kicad_sch | |
| **Title:** | |
| Size: A2 | Date: |
| KiCad E.D.A. 10.0.4 | **Rev:** |
| | Id: 3/22 |

## 2x VL53L7CX

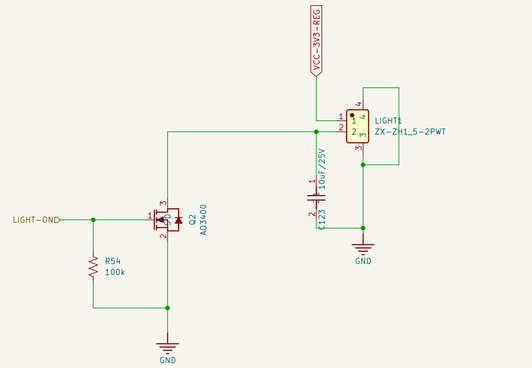



## 2 IR RECEIVERS



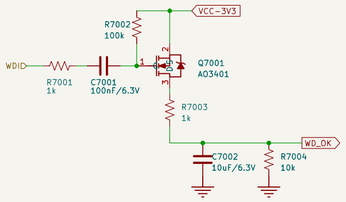

## IR LIGHT

https://thepihut.com/products/850nm-ir-led-modules?srsltid=AfmBOorZOhl4HyIQHd5JbNfaA7bqJMbzrmYObn7Mm12pD_mKEoIREOnX



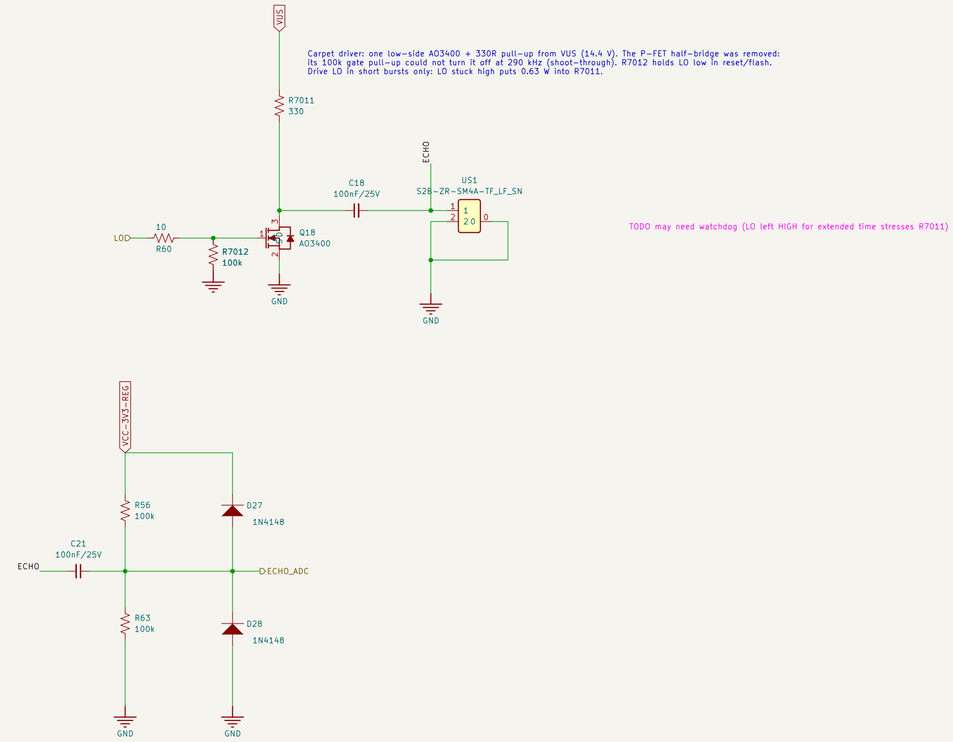

| | | |
|---|---|---|
| Sheet: /FRONT SENSORS/ | | |
| File: Front-Sensors.kicad_sch | | |
| **Title:** | | |
| Size: A2 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | | Id: 4/22 |

## RC CHARGE-PUMP WATCHDOG: WD_OK gates VM-VBAT, VM-5V-LIDAR and VM-5V-WATER-PUMP



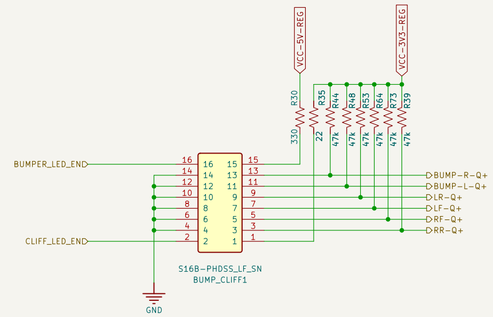

Firmware toggles WDI in software (50–1000 Hz, ~50% duty) only while the control loop is healthy.
Never drive WDI from a timer/PWM: a timer keeps running after the CPU hangs.
Each falling edge on WDI pulls the Q7001 gate low through C7001 and tops up C7002 via R7003.
WD_OK ~2.7 V while toggling. WDI stuck high, low or floating: <1 V in ~100 ms, <0.4 V in ~190 ms
(10 uF nominal; DC-bias derating shortens this). WD_OK starts at 0 V at power-up (fail-safe).
WD_OK drives N-FET gates only (14.4 V rail switch, water pump, LIDAR incl. laser): no resistive loads.
WDI pin: high-Z at reset, no pull-down. All parts reuse existing BOM lines.

| | | |
|---|---|---|
| Sheet: /WATCHDOG/ | | |
| File: WATCHDOG.kicad_sch | | |
| **Title:** | | |
| Size: A2 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | | Id: 5/22 |




| | | |
|---|---|---|
| Sheet: /CARPET-SENSOR/ | | |
| File: Carpet-sensor.kicad_sch | | |
| **Title:** | | |
| Size: A2 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | | Id: 6/22 |

# ANTI FALL SENSORs

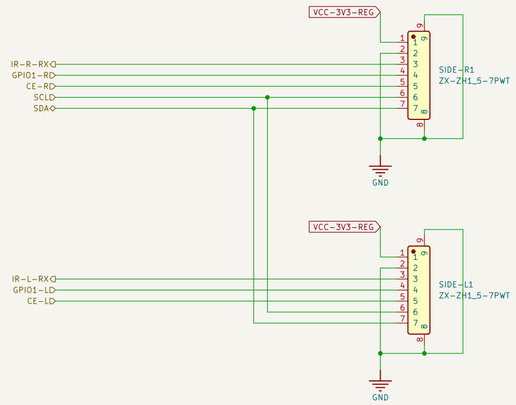



Sheet: /CLIFF/
File: CLIFF.kicad_sch

**Title:**

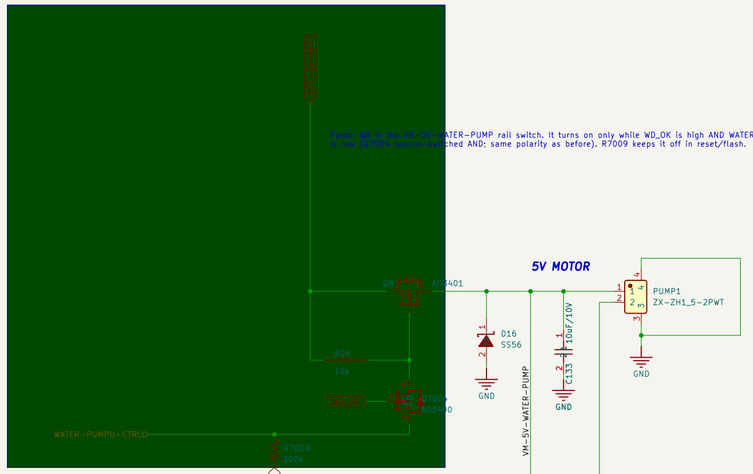
Size: A2 | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 19/22

## *SIDE PROXI IR SENSORs*



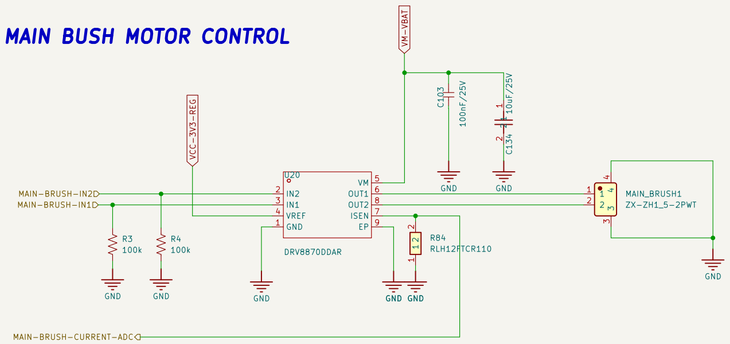

| | | |
|---|---|---|
| Sheet: /SIDE-PROXI/ | | |
| File: SIDE-PROXIMITY-IR-SENSOR .kicad_sch | | |
| ***Title:*** | | |
| Size: A2 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | | Id: 20/22 |

# WATER PUMP MOTOR CONTROL



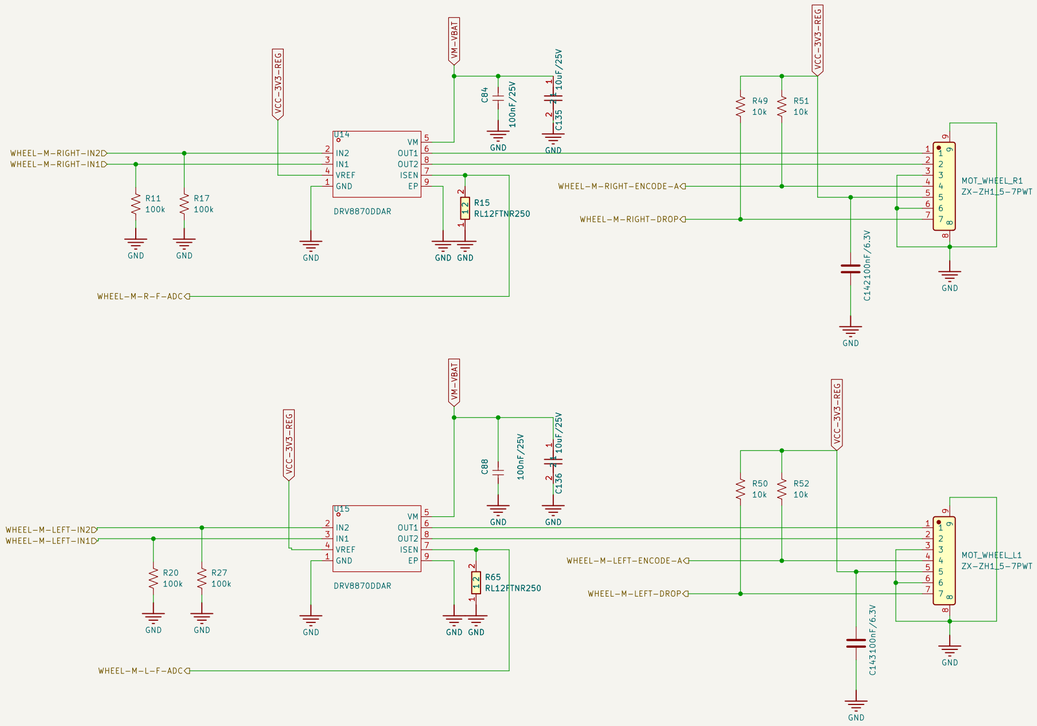

| | |
|---|---|
| Sheet: /WATER-PUMP/ | |
| File: WATER-PUMP.kicad_sch | |
| **Title:** | |
| Size: A2 | Date: |
| KiCad E.D.A. 10.0.4 | **Rev:** |
| | Id: 21/22 |

# MAIN BUSH MOTOR CONTROL




| | | |
|---|---|---|
| Sheet: /MAIN-BRUSH MOTOR/ | | |
| File: MAIN-BRUSH-MOTOR .kicad_sch | | |
| **Title:** | | |
| Size: A2 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | | Id: 22/22 |

# WHEEL MOTOR CONTROL

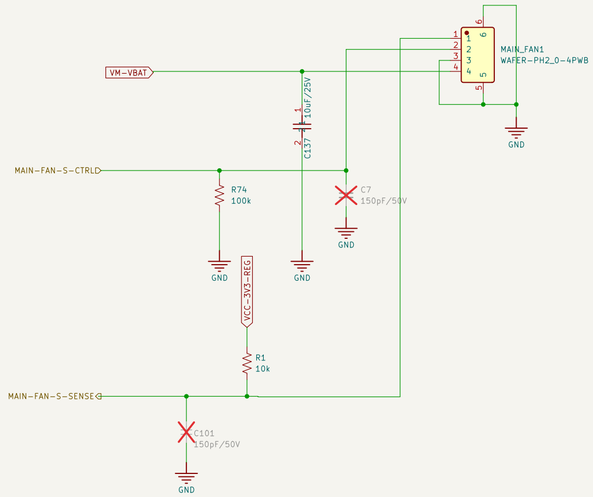



Sheet: /WHEEL MOTOR/
File: WHEEL-MOTORs .kicad_sch

**Title:**

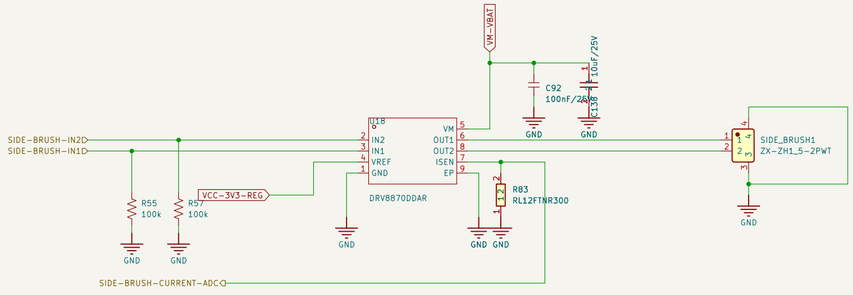
Size: A2 | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 23/22

# MAIN FAN CONTROL




| | | |
|---|---|---|
| Sheet: /MAIN-FAN/ | | |
| File: MAIN-FAN .kicad_sch | | |
| **Title:** | | |
| Size: A2 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | | Id: 24/22 |

# SIDE-BRUSH MOTOR CONTROL



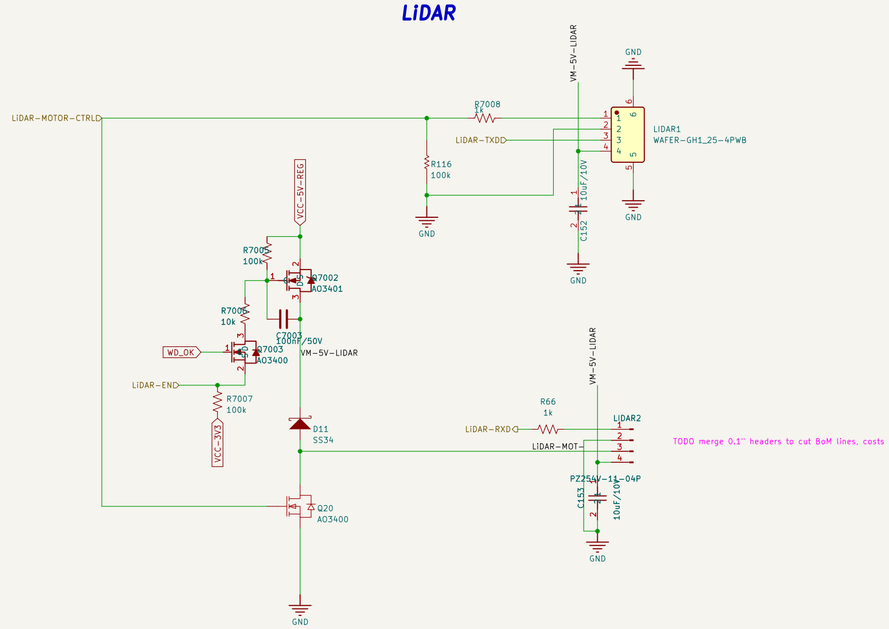

Sheet: /SIDE-BRUSH/
File: SIDE-BRUSH-MOTORs .kicad_sch

**Title:**

| Size: A2 | Date: | **Rev:** |
|---|---|---|
| KiCad E.D.A. 10.0.4 | | Id: 25/22 |

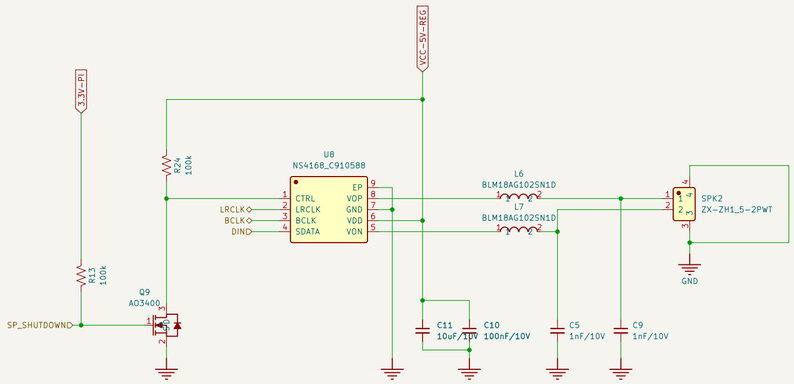
VM-5V-LIDAR powers the whole LiDAR (laser included) for any model. On only while WD_OK is high
AND LiDAR-EN is low (Q7003 source-switched AND). R7006/C7003: ~1.5 ms soft start, ~30 ms soft stop.
LIDAR2: 0.1in header for other LiDARs (pin 2 now GND).

# LiDAR




| | | |
|---|---|---|
| Sheet: /LiDAR/ | | |
| File: LiDAR.kicad_sch | | |
| **Title:** | | |
| Size: A2 | Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | | Id: 26/22 |

# SPEAKER



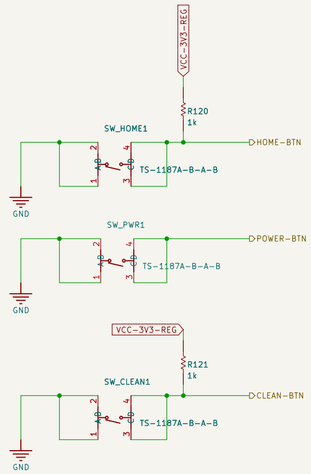

Sheet: /SPEAKER/
File: SPEAKER.kicad_sch

**Title:**

| Size: A2 | Date: | **Rev:** |
|---|---|---|
| KiCad E.D.A. 10.0.4 | | Id: 27/22 |

# LEDs AND BUTTONs



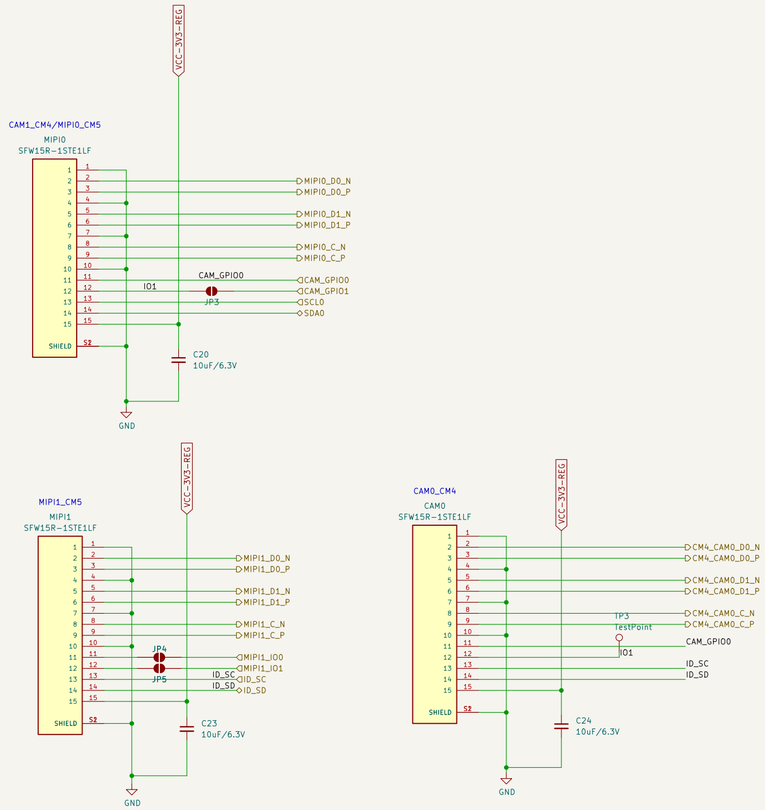




Sheet: /BUTTON-LED/
File: BUTTON-LEDs .kicad_sch

**Title:**

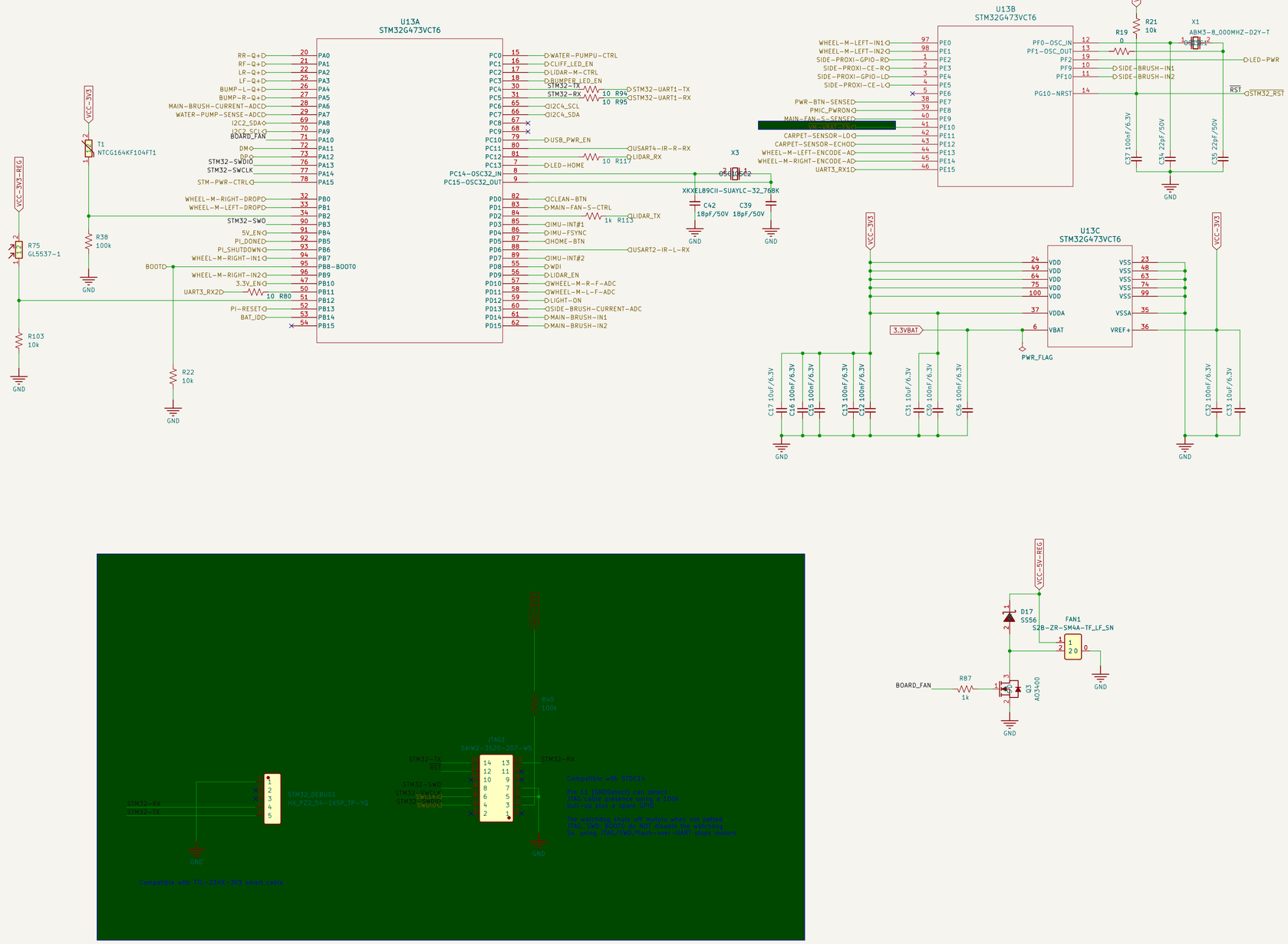
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| KiCad E.D.A. 10.0.4 | | Id: 28/22 |

# MIPI Cameras
# 2 lanes / camera




| | |
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| File: MIPI-CAMERA.kicad_sch | |
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| KiCad E.D.A. 10.0.4 | **Rev:** Id: 29/22 |

# MCU STM32G473







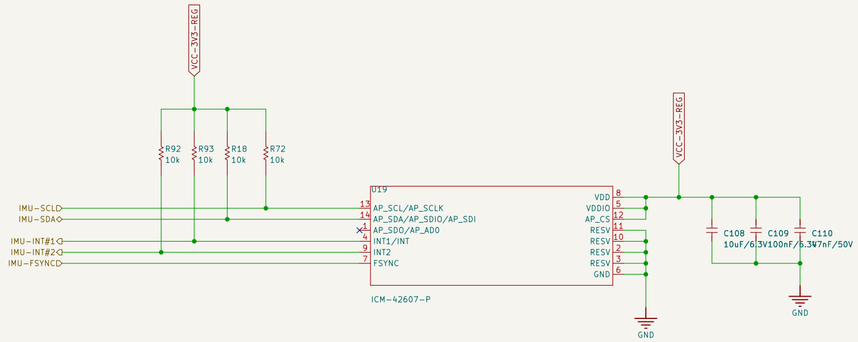



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# IMU-ICM-42607-P

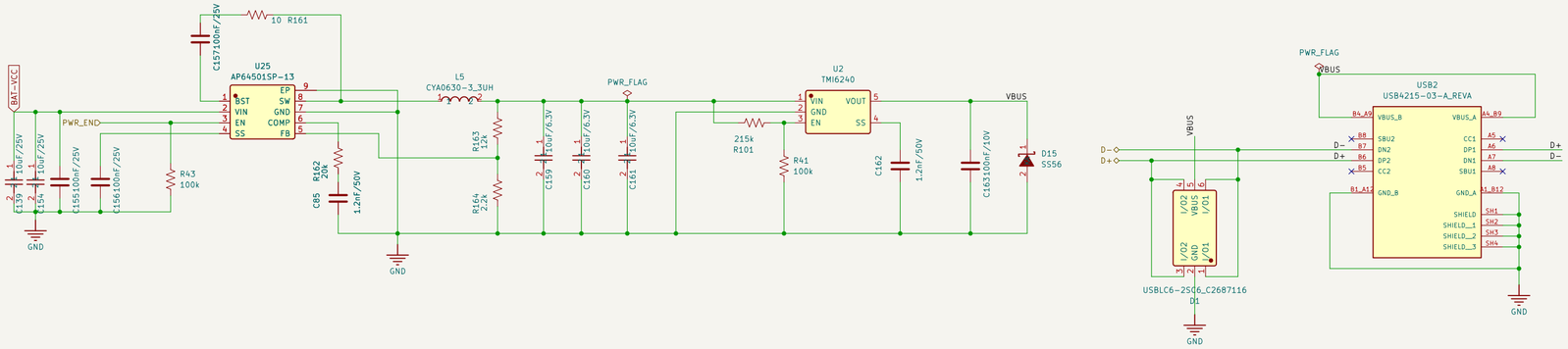



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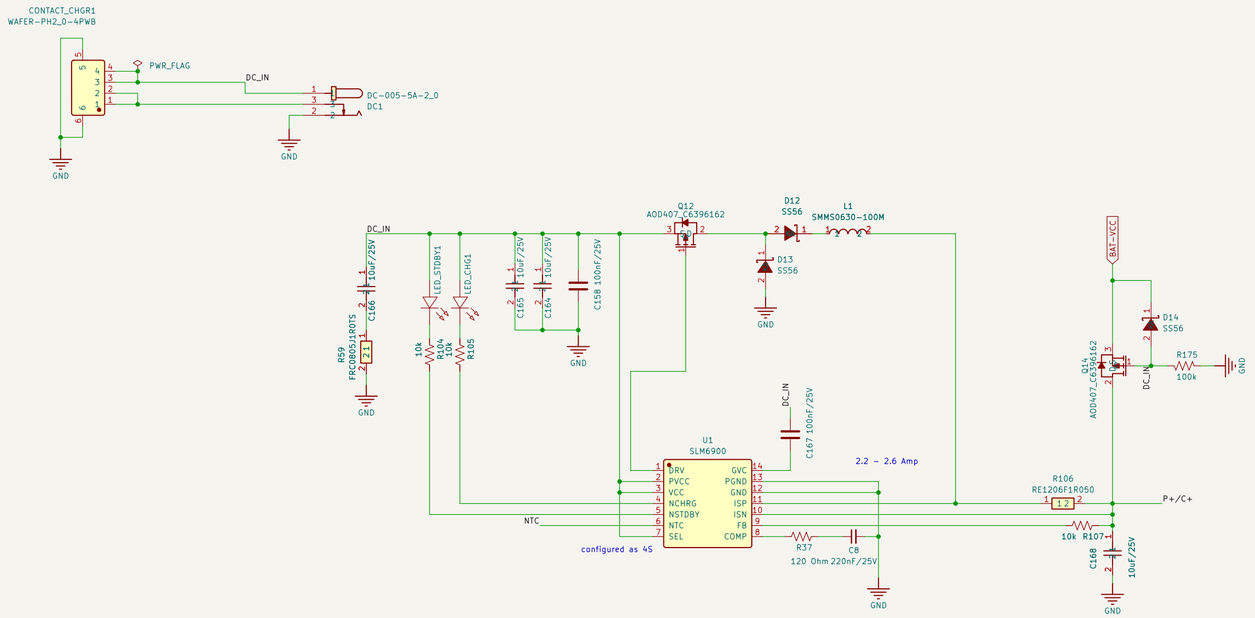
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| KiCad E.D.A. 10.0.4 | | Id: 32/22 |

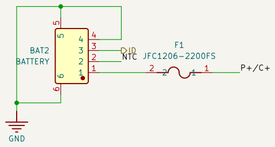
## USB-C



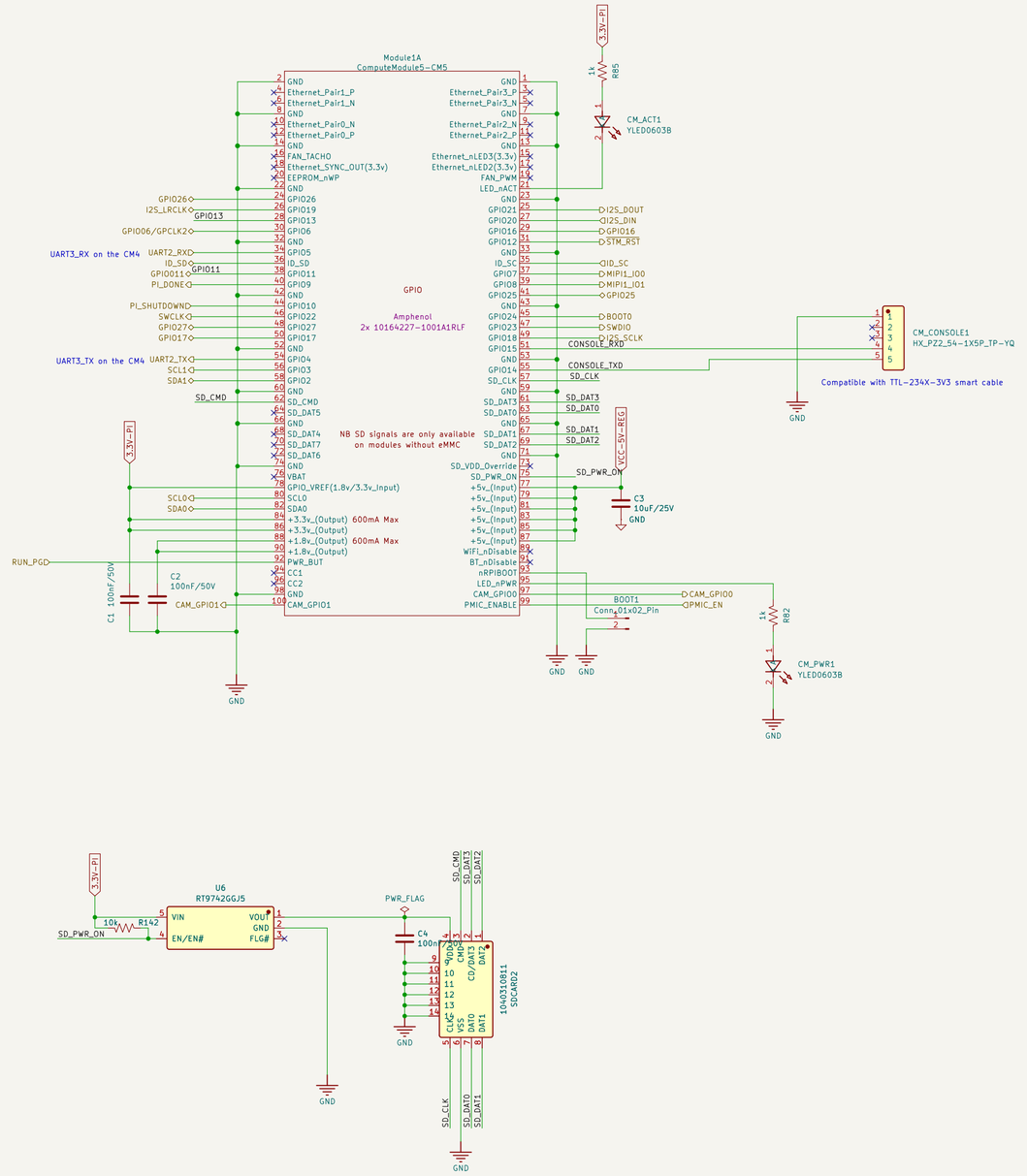

## BATTERY CHARGER




## BATTERY CONNECTOR



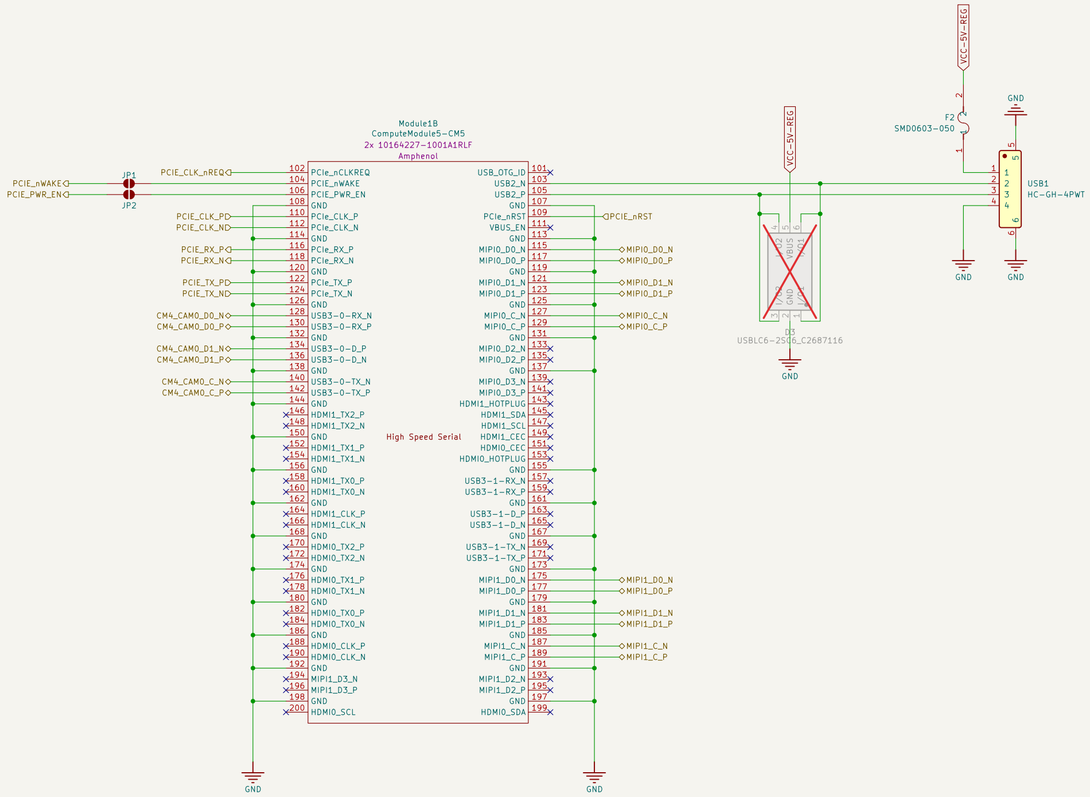

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| KiCad E.D.A. 10.0.4 | | Id: 34/22 |




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